

Contact

abdulbasittonmoy09@gmail.com

www.linkedin.com/in/abdul-basit-tonmoy (LinkedIn)

github.com/abtonmoy (Other)

Top Skills

Robotics

Arduino

C++

Languages

Urdu (Limited Working)

Bangali (Native or Bilingual)

Hindi (Limited Working)

English (Native or Bilingual)

Honors-Awards

Presidential International
Scholarship

Dean's List

Dean's List

Dean's List

J. Crawford Polley Mathematical
Writing Prize

Publications

Advanced Facial Emotion
Classification with 135 Classes
for Enhanced Cybersecurity
Applications

Abdul Basit Tonmoy

ML Researcher || CS & Mathematics @Wabash College ||

Presidential International Scholar

Crawfordsville, Indiana, United States

Summary

I'm an ML researcher and builder working at the intersection of representation learning, multimodal systems, and applied machine learning. I split my time between research and shipping production systems. On the research side: SemanticSentry, a study on the limits of geometric similarity metrics in interpretability (submitted to BlackboxNLP 2026); AdaFrame, a cost-efficient video understanding pipeline for vision-language models (under review at ACM Multimedia 2026); and a 135-class facial emotion recognition system for cybersecurity applications (IEEE ICAIC 2025). I care about experimental rigor: matched-magnitude controls, multi-seed evaluation, and pre-registration whenever the result matters. On the building side: I'm currently co-founding a stealth startup. Past work spans bioinformatics, ad intelligence (CalHacks-12 winner), autonomous robotics, and full-stack data platforms. Outside of work, I make and drink a lot of coffee. It's the through-line for most of my best ideas and most of my worst sleep. I also hike whenever I can get away from a screen. Always happy to talk about representation learning, multimodal systems, interpretability, coffee, trails, or applied ML. Reach out.

Experience

Wabash College

2 years 5 months

QSC Mentor (Math & CS)

January 2026 - Present (5 months)

Crawfordsville, IN

- Tutored Freshman and Sophomore Math and CS students with Homework and exam preparations.
- Coordinated Introductory Class Materials with the Professor
- Helped the Professor update the syllabus based on the student experience in the tutorial

Resident Advisor

August 2024 - Present (1 year 10 months)

Crawfordsville, Indiana, United States

- Fostered a positive, inclusive living environment, mediated conflicts, and organized community-building events for residents.
- Ensured adherence to housing policies, provided academic and resource support, and acted as a first responder during emergencies

Machine Learning Researcher

January 2026 - April 2026 (4 months)

- Built AdaFrame, a three-tier deduplication cascade (perceptual hash → LPIPS → CLIP clustering) with adaptive per-video frame budgets derived from SVD-based Intrinsic Semantic Dimensionality.
- Evaluated on 500 videos from the Pitt Ads Dataset: cut VLM inference cost by 75% and frame count by 48% vs. uniform sampling, while improving topic accuracy to 86.0% (+1.6pp) and sentiment accuracy to 79.4% (+7.5pp).
- Validated provider-agnostic design across Gemini 2.5 Flash and Claude 3 Haiku (<1pp variation across all metrics), and verified ISD strongly correlates with editing complexity ($r = 0.859$, $p < 0.001$).

(Paper under review at ACM Multimedia 2026)

Summer Research Assistant (Robotics)

May 2025 - August 2025 (4 months)

Crawfordsville, IN

Research Intern | Wabash College Department of Mathematics and Computer Science | Summer 2025 |

Supervisor: Dr. Qixin Deng

- Architected hybrid autonomous navigation system with GPS waypoints, computer vision, and LiDAR integration, achieving 97% obstacle detection accuracy and 2-meter precision localization Implemented multi-sensor fusion algorithms with pixel-to-angle mapping and emergency protocols for real-time obstacle avoidance in dynamic environments
- Developed adaptive control systems with serial communication protocols, enabling autonomous navigation under varying terrain and weather conditions

Machine Learning Research Intern

August 2024 - April 2025 (9 months)

- Built a U-Net + ResNet model for facial emotion recognition across 135 classes; presented at the 4th IEEE Conference on AI in Cybersecurity 2025.
- Developed a dataset expansion strategy to handle class imbalance across the 135 classes.

- Worked with cybersecurity and HCI collaborators on applications in behavioral threat detection and adaptive authentication. <https://ieeexplore.ieee.org/document/10848788>

Data Analyst for Auxiliary Services

January 2024 - March 2025 (1 year 3 months)

Crawfordsville, IN

- Built automated fleet analysis pipeline in Python processing 15+ years of vehicle data, implementing statistical models and Excel integration for resource optimization
- Generated \$20,160 in annual cost savings and \$3,200 per-vehicle revenue increase through data-driven acquisition strategy recommendations and predictive modeling

Teaching Assistant (Physics)

September 2024 - February 2025 (6 months)

Crawfordsville, IN

- Assisted with laboratory sessions and problem-solving workshops for introductory physics courses, providing one-on-one guidance to students on experimental techniques, data analysis, and conceptual understanding
- Graded assignments and lab reports, providing constructive feedback to support student learning and mastery of physics principles

Summer Software Engineering Intern

May 2024 - July 2024 (3 months)

Crawfordsville, Indiana, United States

- Developed an interactive analytics dashboard leveraging React, Redux, and Material-UI, enabling real-time data analytics and visualizations and improving data accessibility using the last 20 years of data.
- Implemented secure authentication mechanisms and advanced search functionalities, reducing user search time by 30% and enhancing overall user experience.
- Optimized backend data handling using MongoDB and Node.js, increasing data processing efficiency by 40% and ensuring scalability for future growth.

Compass Group

Analyst

September 2025 - January 2026 (5 months)

Crawfordsville, IN

- Deployed OCR pipeline to automate receipt-to-Excel data ingestion, eliminating manual entry and accelerating financial processing by 10h/week

Notre Dame Information Technology Club

2 years 6 months

President

September 2022 - May 2023 (9 months)

Dhaka, Bangladesh

Department of Archive, Workshop, and Seminar

Event Coordinator

March 2021 - August 2022 (1 year 6 months)

Bangladesh

Group Representative

December 2020 - February 2021 (3 months)

Bangladesh

Education

Wabash College

Bachelor of Arts - BA, Computer Science · (August 2023 - May 2027)

Wabash College

Bachelor of Arts - BA, Mathematics · (August 2023 - May 2027)

Notre Dame College

High School Diploma, Science · (September 2020 - February 2023)

Chandpur Hasan Ali Government High School

High School Diploma, Science · (January 2015 - July 2020)